



AEO “Nazarbayev Intellectual Schools”

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CHEMISTRY

Grade 12

Paper 1

May 2024

1 hour

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Ruler
 Data Booklet

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue, or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators may be used.

The total number of marks for this paper is 40.

Answer all questions in English.

For Examiner's Use

Section A

Section B

Total

This document consists of **18** printed and **2** blank pages.

Section A

For
Examiner's
Use

For each question, there are four possible answers **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct by putting a tick (✓) in the correct box.

- 1 Oxidation and reduction are defined in terms of oxygen, hydrogen, and electron transfer. Which of the following statements is correct?

A Reduction is the loss of hydrogen.

B Oxidation is the gain of oxygen.

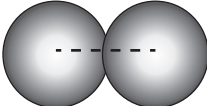
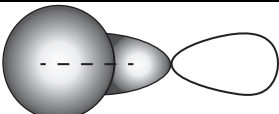
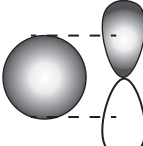
C Reduction is the loss of electrons.

D Oxidation is the gain of electrons.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 2 Which overlapping of orbitals is involved in a PCl_3 molecule?

A	
B	
C	
D	

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

3 There are four stages of a mass spectrometer.

- 1 ionisation
- 2 detection
- 3 deflection
- 4 acceleration

What is the correct order for the stages in the mass spectrometer?

- A** 1; 2; 4; 2
- B** 1; 3; 4; 2
- C** 1; 4; 2; 3
- D** 1; 4; 3; 2

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

4 The table shows the properties of some elements and their compounds.

	XY	Z	ZQ	X
State (s.t.p.)	solid	solid	gas	solid
Melting point	high	very high	low	high
Water solubility	soluble	insoluble	insoluble	insoluble
Electrical conductivity	good in aqueous solutions	poor	poor	good

Determine the elements **X**, **Y**, **Z**, and **Q**.

- A** **X** – Li, **Y** – F, **Z** – C, **Q** – H
- B** **X** – B, **Y** – O, **Z** – H, **Q** – Li
- C** **X** – C, **Y** – F, **Z** – Li, **Q** – H
- D** **X** – Li, **Y** – F, **Z** – C, **Q** – O

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 5 Use of the Data Booklet is relevant to this question.

A 13.8 g sample of a compound containing only nitrogen and oxygen produced 4.2 g of nitrogen upon decomposition. What is its empirical formula?

- A NO
B NO₂
C N₂O₄
D N₂O₅

A ☐ B ☐ C ☐ D ☐

[1]

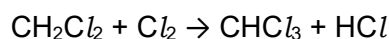
- 6 What is the **strongest** intermolecular force present for each of the following molecules?

	Methylamine (CH ₃ NH ₂)	Carbon dioxide (CO ₂)	Hydrogen chloride (HCl)
A	hydrogen bonding	induced dipole-dipole forces	permanent dipole-dipole forces
B	hydrogen bonding	permanent dipole-dipole forces	permanent dipole-dipole forces
C	permanent dipole-dipole forces	induced dipole-dipole forces	permanent dipole-dipole forces
D	permanent dipole-dipole forces	induced dipole-dipole forces	hydrogen bonding

A ☐ B ☐ C ☐ D ☐

[1]

- 7 Trichloromethane is formed when dichloromethane reacts with chlorine.



The mechanism is a free-radical substitution reaction.

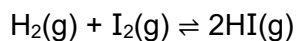
Choose the equation for the **first** propagation step in the mechanism.

- A $\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$
B $\text{Cl}_2 + \cdot\text{CHCl}_2 \rightarrow \text{CHCl}_3 + \text{Cl}\cdot$
C $2\cdot\text{CHCl}_2 \rightarrow \text{CHCl}_2\text{CHCl}_2$
D $\text{Cl}\cdot + \text{CH}_2\text{Cl}_2 \rightarrow \cdot\text{CHCl}_2 + \text{HCl}$

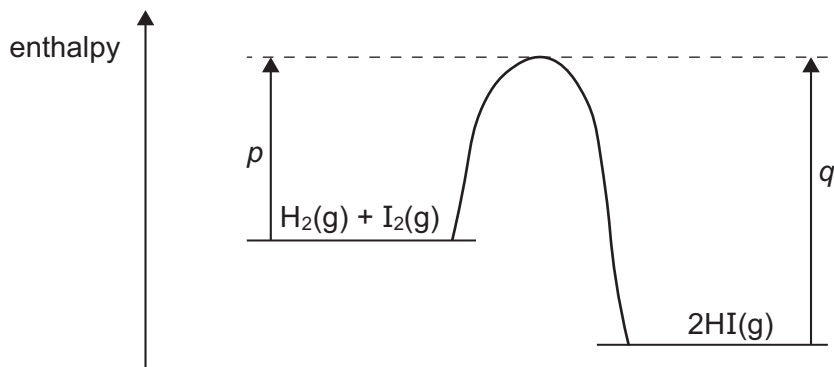
A ☐ B ☐ C ☐ D ☐

[1]

- 8 The equation below shows the equilibrium reaction when hydrogen and iodine react to form hydrogen iodide.



The curve in the diagram below illustrates the reaction profile for this equilibrium reaction without a catalyst.



Which row of the table correctly describes the reaction?

	value for the activation energy for the forward reaction	value for the overall enthalpy change for the forward reaction
A	p	– (q – p)
B	q	– (p – q)
C	p	+ (q – p)
D	q	+ (p – q)

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 9 There are four isomeric alcohols with the molecular formula $\text{C}_4\text{H}_{10}\text{O}$. One isomer, alcohol **Q**, does not react with acidified potassium dichromate(VI) solution.

Name the alcohol **Q**.

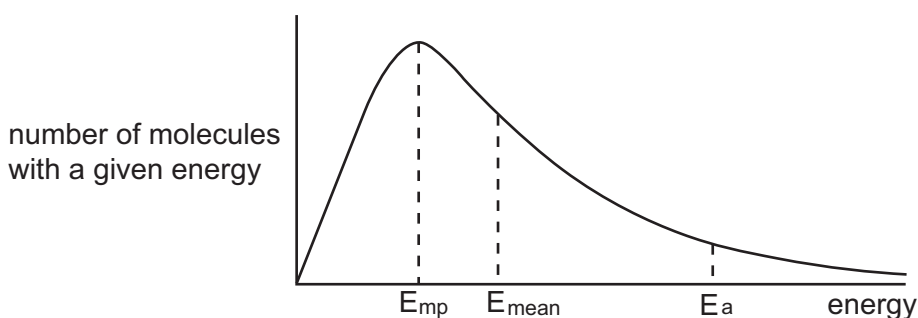
- A** butan-1-ol
B butan-2-ol
C 2-methylpropan-1-ol
D 2-methylpropan-2-ol

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 10 The diagram below shows a Maxwell–Boltzmann distribution for a sample of gas at a fixed temperature.

For
Examiner's
Use



Where:

E_{mp} the value of the most probable energy
 E_{mean} the value of the mean energy of the particles
 E_a activation energy

What happens to the values on the curve as the temperature of the gas increases?

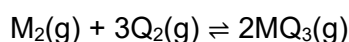
	E_{mp}	E_{mean}	the number of molecules with energy greater than E_a
A	increases	increases	remains the same
B	decreases	increases	remains the same
C	increases	increases	increases
D	decreases	increases	increases

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 11 M and Q are chemical elements.

Reactants $M_2(g)$ and $Q_2(g)$ react exothermically to produce a compound $MQ_3(g)$, as shown by the equation.



The dynamic equilibrium was established at temperature T in a closed container.

Which row of the table correctly describes the following effects on the equilibrium?

	increasing the temperature	increasing the volume of the container
A	amount of $M_2(g)$ increase	amount of $MQ_3(g)$ decrease
B	amount of $M_2(g)$ decrease	amount of $MQ_3(g)$ increase
C	amount of $Q_2(g)$ increase	amount of $MQ_3(g)$ increase
D	amount of $Q_2(g)$ decrease	amount of $MQ_3(g)$ decrease

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

12 Use of the Data Booklet is relevant to this question.

Which electronic configuration is correct for an isotope with the same number of electrons as in an atom of ^{20}Ne ?

- A $[\text{He}] 2s^2 2p^4$
 B $[\text{Ne}] 3s^2 3p^4$
 C $[\text{He}] 2s^2 2p^6$
 D $[\text{Ar}] 4s^2$

A ☐ B ☐ C ☐ D ☐

[1]

13 The table contains some standard electrode potential data.

electrode half-equation	$E^\ominus / \text{V}$
$\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$	+ 0.80
$\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$	− 0.44

A cell is made by connecting Fe^{2+}/Fe and Ag^+/Ag half-cells with a salt bridge.

Identify the cell potential **and** the direction of the electrons' flow.

- A 1.24 V, from Fe^{2+}/Fe to Ag^+/Ag
 B 0.36 V, from Fe^{2+}/Fe to Ag^+/Ag
 C 0.36 V, from Ag^+/Ag to Fe^{2+}/Fe
 D 1.24 V, from Ag^+/Ag to Fe^{2+}/Fe

A ☐ B ☐ C ☐ D ☐

[1]

14 Unsaturated fatty acids can be hydrogenated to saturated fatty acids in the presence of Ni catalyst.

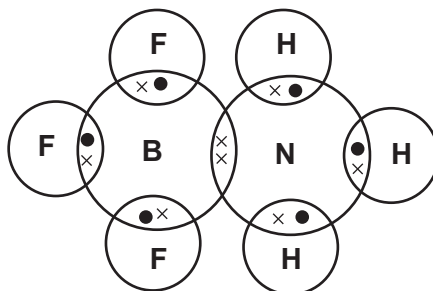
Identify the reaction mechanism.

- A nucleophilic addition
 B nucleophilic substitution
 C electrophilic addition
 D electrophilic substitution

A ☐ B ☐ C ☐ D ☐

[1]

- 15 The shown dot and cross diagram illustrates the structure of the compound BF_3NH_3 .



Choose the correct statement according to the given diagram.

- A N—B is a dative bond. BF_3 acts as a Lewis acid.
 B N—B is a covalent bond. NH_3 acts as a Lewis acid.
 C N—H is a dative bond. BF_3 acts as a Lewis base.
 D B—F is a covalent bond. NH_3 acts as a Lewis acid.

A ☐ B ☐ C ☐ D ☐

[1]

- 16 Most of the elements in Group 14 form compounds with chlorine. These compounds are called tetrachlorides.

Which of the following statements is **NOT** true?

- A The shape of the tetrachlorides is tetrahedral.
 B The Group 14 tetrachlorides have simple covalent molecules.
 C The Group 14 tetrachlorides have low boiling points.
 D The bond angle in carbon tetrachloride is 102° .

A ☐ B ☐ C ☐ D ☐

[1]

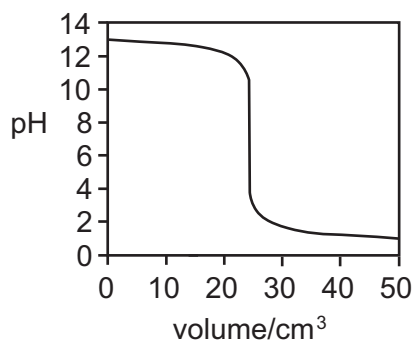
- 17 In which of these chemical processes does the nitric acid, HNO_3 , behave as a Bronsted-Lowry base?

- A $\text{HNO}_3(\text{aq}) \rightarrow \text{NO}_3^-(\text{aq}) + \text{H}^+(\text{aq})$
 B $\text{H}_2\text{SO}_4(\text{aq}) + \text{HNO}_3(\text{aq}) \rightarrow \text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + \text{NO}_2^+(\text{aq})$
 C $\text{NH}_3(\text{aq}) + \text{HNO}_3(\text{aq}) \rightarrow \text{NH}_4\text{NO}_3(\text{aq})$
 D $\text{HNO}_3(\text{aq}) + \text{KOH}(\text{aq}) \rightarrow \text{KNO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$

A ☐ B ☐ C ☐ D ☐

[1]

18 A pH curve is shown below.



The table shows information about some acid-base indicators.

indicator	pH range	lower pH colour	higher pH colour
Naphthyl red	3.7–5.0	red	yellow
Cresol purple	7.6–9.2	yellow	purple

All solutions have concentrations of 0.1 mol dm^{-3} .

Choose reagents and the indicator for the titration curve.

	reagents	indicator and colour change at the end point of the titration
A	sodium hydroxide to 25 cm^3 of ethanoic acid	naphthyl red, from yellow to red
B	hydrochloric acid to 25 cm^3 of potassium hydroxide	naphthyl red, from red to yellow
C	potassium hydroxide to 25 cm^3 of hydrochloric acid	cresol purple, from yellow to purple
D	hydrochloric acid to 25 cm^3 of potassium hydroxide	cresol purple, from purple to yellow

A ☐ B ☐ C ☐ D ☐

[1]

19 $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ complex ions are pink coloured.

Identify the factor that **DOES NOT** affect the colour of a transition metal complex ion.

- A nature of the ligand
- B oxidation state of the metal
- C shape of the complex ion
- D concentration of the complex ion

A ☐ B ☐ C ☐ D ☐

[1]

20 Aqueous copper(II) ions $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ are acidic and react with a base.

When a few drops of $\text{NH}_3(\text{aq})$ are added to copper(II) ions, a blue precipitate is formed, while in excess $\text{NH}_3(\text{aq})$ the blue precipitate is dissolved and becomes a royal blue aqueous solution.

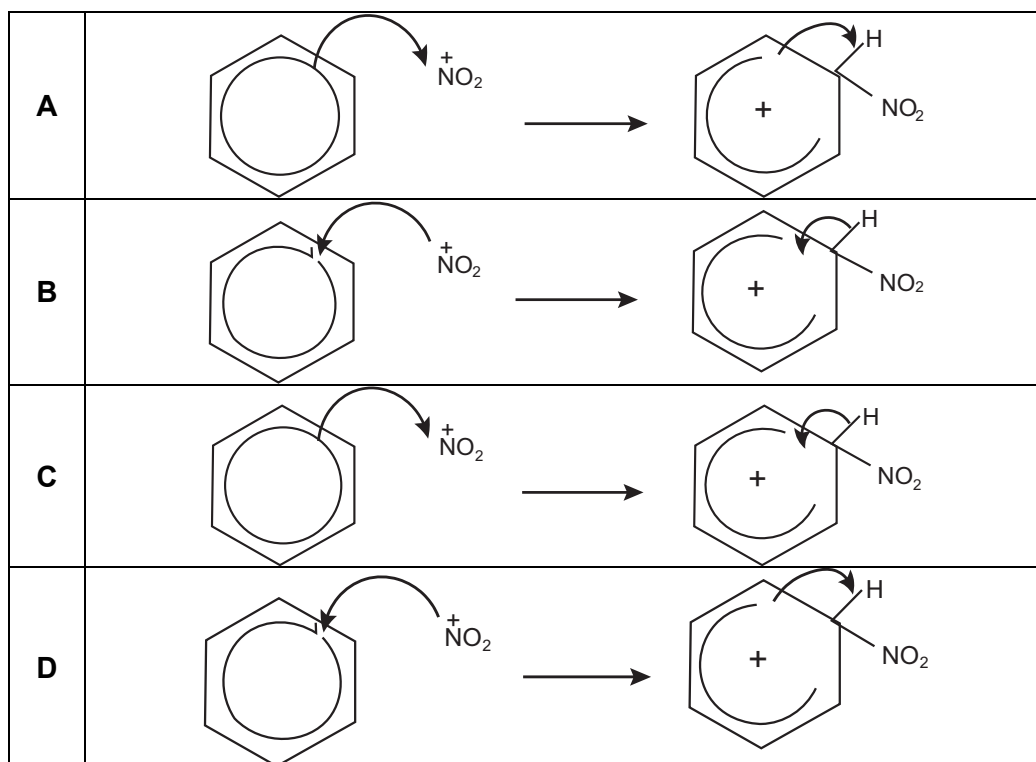
Identify the correctly balanced equation for the formation of a royal blue aqueous solution.

- A $[\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4](\text{s}) + 2\text{NH}_3(\text{aq}) \rightarrow [\text{Cu}(\text{NH}_3)_2(\text{H}_2\text{O})_4]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq})$
- B $[\text{Cu}(\text{OH})(\text{H}_2\text{O})_5]^+(\text{aq}) + \text{NH}_3(\text{aq}) \rightarrow [\text{Cu}(\text{NH}_3)(\text{H}_2\text{O})_5]^{2+}(\text{aq}) + \text{OH}^-(\text{aq})$
- C $[\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4](\text{s}) + 4\text{NH}_3(\text{aq}) \rightarrow [\text{Cu}(\text{NH}_3)_4(\text{H}_2\text{O})_2]^{2+}(\text{aq}) + 2\text{OH}^-(\text{aq}) + 2\text{H}_2\text{O}(\text{l})$
- D $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq}) + 2\text{NH}_3(\text{aq}) \rightarrow [\text{Cu}(\text{OH})_2(\text{H}_2\text{O})_4](\text{s}) + 2\text{NH}_4^+(\text{aq})$

A ☐ B ☐ C ☐ D ☐

[1]

21 Which of the following equations shows the mechanism for the reaction between benzene and an electrophile?



A ☐ B ☐ C ☐ D ☐

[1]

22 Compounds **Q** and **S** are isomers with the molecular formula $C_4H_{10}O$.

Experiments with **Q** and **S** were carried out separately and the following results were obtained.

Compound **Q** + [O] → Compound **N**.

Compound **N** + Tollen's reagent → Silver mirror formed.

Compound **S** + [O] → No visible change.

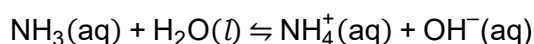
Identify formulas of compounds **Q**, **S**, and **N**.

	Compound Q	Compound N	Compound S
A	$CH_3CH(CH_3)CH_2OH$	$CH_3CH(CH_3)COH$	$CH_3C(OH)(CH_3)CH_3$
B	$CH_3CH_2CH_2CH_2OH$	$CH_3CH_2CH_2COOH$	$CH_3C(OH)(CH_3)CH_3$
C	$CH_3CH(CH_3)CH_2OH$	$CH_3CH(CH_3)COH$	$CH_3CH(OH)CH_2CH_3$
D	$CH_3CH_2CH_2CH_2OH$	$CH_3C(O)CH_2CH_2OH$	$CH_3C(OH)(CH_3)CH_3$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

23 A basic buffer solution is a mixture of a weak base and its conjugate strong acid. At equilibrium this system can be represented by:



Predict the change in equilibrium when a small amount of acid is added to this buffer solution.

- A** concentration of ammonia increases
- B** concentration of hydroxide ions decreases
- C** concentration of ammonium ions decreases
- D** concentration of reactants remains constant

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

24 The structures of four isomers, with the molecular formula $C_5H_{10}O$ are shown below. Which of the following has **both** optical and E–Z isomers?

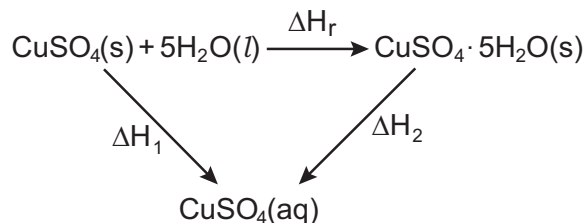
- A** $CH_3-CH=CH-CH_2-CH_2OH$
- B** $CH_3-CH=CH-CH(OH)-CH_3$
- C** $CH_2=CH-CH_2-CH(OH)-CH_3$
- D** $CH_2=CH-CH_2-CH_2-CH_2OH$

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 25 The enthalpy change for the reaction to form hydrated copper sulphate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, crystals cannot be measured directly.

Determine the expression used to calculate the enthalpy change for the reaction.



- A $\Delta H_1 - \Delta H_2$
 B $\Delta H_1 + \Delta H_2$
 C $-\Delta H_1 - \Delta H_2$
 D $-\Delta H_1 + \Delta H_2$

A ☐ B ☐ C ☐ D ☐

[1]

- 26 The Valence Shell Electron Pair Repulsion (VSEPR) theory is used to predict the shapes of molecules.

Which of the following statements about CH_3^- is correct according to the VSEPR theory?

- A the number of lone pairs = 0
 B the number of bond pairs = 1
 C the shape is trigonal pyramidal
 D the bond angle = 120°

A ☐ B ☐ C ☐ D ☐

[1]

- 27 Use the Data Booklet to answer this question.

The tertiary structure of a protein refers to the overall 3D shape of its polypeptide chain.

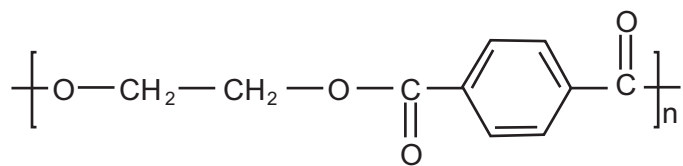
Which of the following side groups of the polypeptide chain is interacted in ionic bonding?

- A Trp and Leu
 B Glu and Arg
 C Tyr and His
 D Cys and Cys

A ☐ B ☐ C ☐ D ☐

[1]

- 28 Terylene, also known as Dacron, is a polyester with ester linkages.



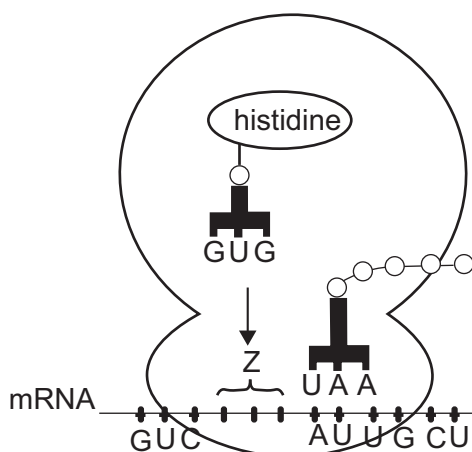
Identify the monomers used to form Terylene.

- A** ethane-1,2-diol and benzene-1,4-dioyl chloride
B ethane-1,2-diol and benzene-1,4-diamine
C ethanol and benzene-1,4-dioic acid
D ethane-1,2-diol and benzoic acid

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 29 The part of the process of protein synthesis is shown below.



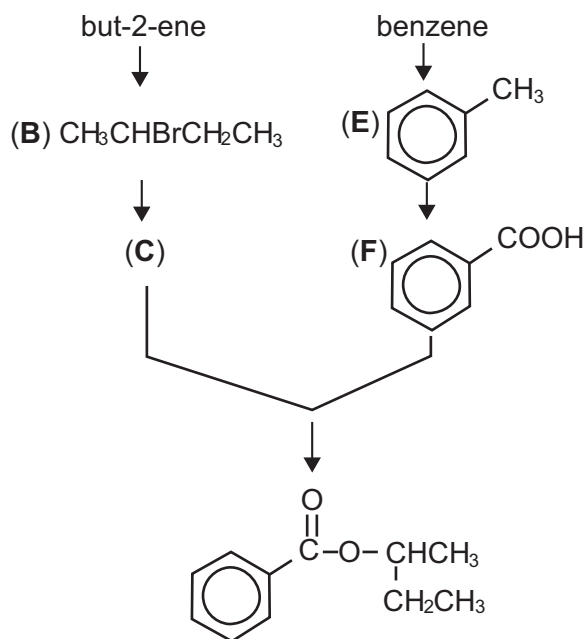
Identify the part of the sequence, labeled Z, of the mRNA codon and DNA codon.

- A** CAC and GUG
B CAC and GTG
C CUC and GUG
D GUG and CAC

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

30 A student carried out the following reaction scheme starting with but-2-ene and benzene.



Which of the following statements is correct regarding the above scheme?

- A But-2-ene undergoes electrophilic substitution reaction.
- B The reagent used to convert (B) to (C) is an alcoholic hydroxide solution.
- C The catalyst required in converting benzene to (E) is AlCl_3 .
- D The product (C) is butan-1-ol.

A ☐ B ☐ C ☐ D ☐

[1]

Section B

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Use

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

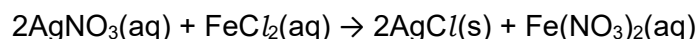
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31** When iron(II) chloride reacts with silver nitrate, iron(II) nitrate and silver chloride are produced.

The balanced equation for this reaction is:



What are the spectator ions in the reaction?

- 1 $\text{Fe}^{2+}(\text{aq})$ and $\text{NO}_3^{-}(\text{aq})$
- 2 $\text{Ag}^{+}(\text{aq})$ and $\text{Cl}^{-}(\text{aq})$
- 3 $\text{Ag}^{+}(\text{aq})$ and $\text{NO}_3^{-}(\text{aq})$

A ☐ B ☐ C ☐ D ☐

[1]

- 32** Down Group 2 the elements demonstrate general trends.

Which of the following is true?

- 1 Atomic radius and first ionization increase.
- 2 The solubility of hydroxides increases and of sulphates decreases.
- 3 Reactivity of the elements increases.

A ☐ B ☐ C ☐ D ☐

[1]

- 33** Ketones react with hydrogen cyanide, HCN, to yield cyanohydrins.

Which of these ketones **DOES NOT** form a racemic mixture in this reaction?

- 1 CH_3COCH_3
- 2 $\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$
- 3 $\text{CH}_3\text{CH}_2\text{COCH}_3$

A ☐ B ☐ C ☐ D ☐

34 Which of the compounds would be oxidized to a carboxylic acid using acidified sodium dichromate(VI)? [1]

- 1 pentan-1-ol and pentanal
- 2 pentanal and pentan-2-ol
- 3 pentan-2-one and pentan-1-ol

A ☐ B ☐ C ☐ D ☐

[1]

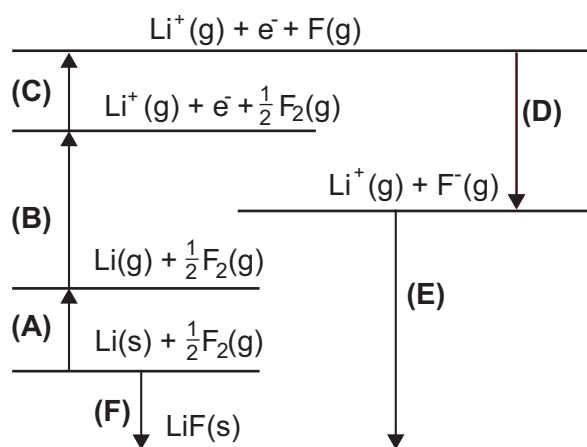
35 In which of the following reactions concentrated sulphuric(VI) acid, H_2SO_4 , acts as an oxidizing agent?

- 1 sulphuric(VI) acid with potassium bromide(aq)
- 2 sulphuric(VI) acid with potassium iodide(aq)
- 3 sulphuric(VI) acid with potassium chloride(aq)

A ☐ B ☐ C ☐ D ☐

[1]

36 The Born-Haber cycle for lithium fluoride is shown.



Which enthalpy change is correctly labeled on the diagram?

- 1 (B) is the first ionization energy of lithium.
- 2 (D) is the first electron affinity.
- 3 (E) is the enthalpy change for the formation of lithium fluoride.

A ☐ B ☐ C ☐ D ☐

[1]

- 37 Fuel cells generate electricity from an electrochemical reaction in which oxygen and hydrogen combine to form water.

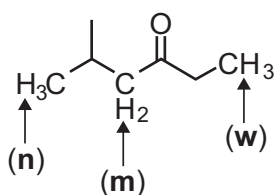
What happens to electrons in a hydrogen-oxygen fuel cell?

- 1 Electrons pass through the electrolyte.
- 2 Hydrogen is oxidised to H^+ ions at the anode.
- 3 Oxygen is reduced at the cathode.

A ☐ B ☐ C ☐ D ☐

[1]

- 38 Identify the splitting patterns expected for the indicated protons in the following structure obtained from proton NMR.

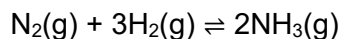


- 1 **n** is a singlet
- 2 **m** is a doublet
- 3 **w** is a triplet

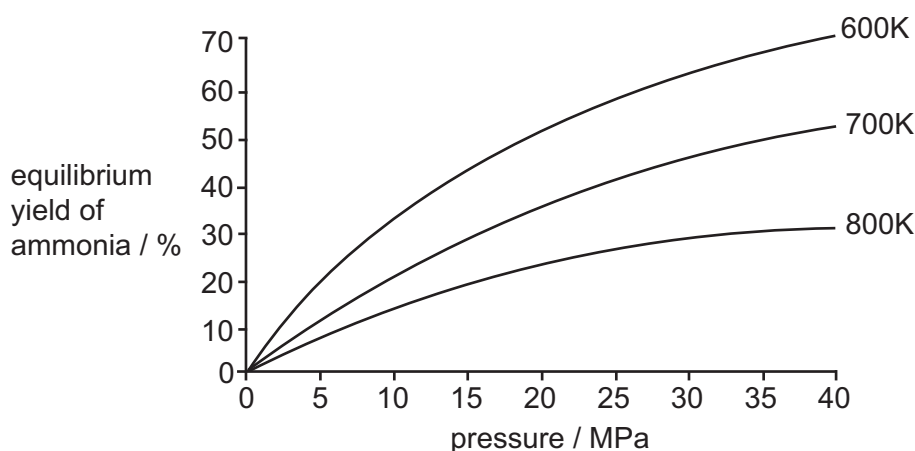
A ☐ B ☐ C ☐ D ☐

[1]

39 Haber process involves a reversible reaction:



The diagram shows how the equilibrium yield of ammonia varies with changes in pressure and temperature.



Use the diagram and Le Chatelier's principle to choose the correct statement(s) about the reversible reaction.

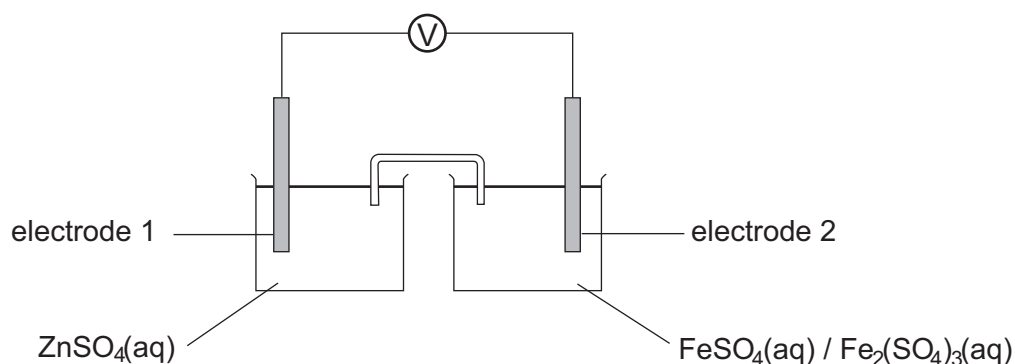
- 1 High yield of ammonia requires low temperature.
- 2 High yield of ammonia requires high pressure.
- 3 The forward reaction is exothermic.

A ☐ B ☐ C ☐ D ☐

[1]

40 Use the Data Booklet to answer this question.

The apparatus below can be used to measure the value of E_{cell} for the reaction.



Which statement(s) about electrodes is(are) correct?

- 1 Electrode 1 is zinc, Zn, and electrode 2 is platinum, Pt.
- 2 Electrons flow from $\text{FeSO}_4(\text{aq})$ to $\text{ZnSO}_4(\text{aq})$.
- 3 Zinc is reduced from 0 to +2.

A ☐ B ☐ C ☐ D ☐

[1]

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